BGP Q&A – Arrcus Prep

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How to use: cover the answer, say yours out loud in 3–4 sentences (what → why → verify → fix), then compare. Questions marked ★ hit the area a past Arrcus candidate got wrong.

1. Sessions and messages

Q1. Neighbour flaps Idle ↔ Active. What does Active mean and what are the causes?

- Active = TCP to port 179 is failing; the router listens and retries on ConnectRetry.
- Causes: no route to peer / wrong IP · update-source mismatch · eBGP to loopback without ebgp-multihop (TTL 1) · TCP 179 blocked (ACL, firewall, CoPP) · MD5 mismatch · peer not configured (RST).
- Wrong remote-as is different: TCP comes up, OPEN rejected → NOTIFICATION → OpenSent/Idle, not Active.

Q2. List the BGP FSM states and message types.

- States: Idle → Connect → Active → OpenSent → OpenConfirm → Established.
- Messages: OPEN (version, AS, hold time, router-ID, capabilities), UPDATE (withdrawn routes, path attributes, NLRI), KEEPALIVE, NOTIFICATION (error, then close), ROUTE-REFRESH.

Q3. How are hold time and keepalive decided?

- Each side proposes a hold time in OPEN; the **lower** value is used. Keepalive = one third of it. Defaults: 60 s keepalive / 180 s hold (Cisco); 0 = no keepalives.
- Hold timer expiry → NOTIFICATION "hold time expired" → session down. Use BFD for sub-second detection.

2. Route propagation ★

Q4 ★. R1 learns 8.8.8.0/24 from eBGP ISP-A (AS 300). R1, R2, R3 are an iBGP full mesh in AS 100. R3 has eBGP to ISP-B (AS 400). Does R3 send it to ISP-B? What changes?

- **Yes** – a best route learned via iBGP **is** advertised to eBGP peers.
- AS_PATH: R3 prepends 100 → ISP-B sees "100 300".
- NEXT_HOP: set to R3 (its address on the ISP-B session).
- LOCAL_PREF: removed (only meaningful inside the AS).
- MED: the MED received from AS 300 is not passed on to AS 400; R3 may set its own.
- Communities: sent only if send-community is configured; **no-export** would stop it here.
- ORIGINATOR_ID / CLUSTER_LIST: removed (iBGP-only).
- Conditions: it must be R3's **best** path and **valid** (next-hop reachable), and outbound policy must permit it.

> **Say it:** *Yes. iBGP-learned best routes go to eBGP peers. On the way out R3 prepends AS 100, sets itself as next-hop, strips LOCAL_PREF and doesn't pass on AS 300's MED. It must be the best and valid path, and no-export or outbound policy would block it.*

Q5 ★. R2 has an iBGP peer R4 that is not peered with R1. Does R4 get 8.8.8.0/24 from R2?

- **No.** iBGP split horizon: a route learned from an iBGP peer is **not** advertised to another iBGP peer.
- Reason: iBGP doesn't change AS_PATH, so AS_PATH loop detection can't work inside the AS. The rule prevents loops but needs a full mesh.
- Fixes: full mesh (add R1–R4), **route reflector** (R2 as RR, R4 client), or **confederation**.

Q6. Summarise the advertisement rules.

Learned from	To eBGP	To iBGP (no RR)	RR → clients	RR → non-clients
eBGP	Yes	Yes	Yes	Yes
iBGP non-client	Yes	**No**	Yes	No
iBGP client	Yes	n/a	Yes	Yes

- Only the best path is advertised (unless ADD-PATH).

Q7. How does a route reflector prevent loops? Does it change the next-hop?

- **ORIGINATOR_ID**: RR sets it to the originating router's ID; a router ignores a route carrying its own ID.
- **CLUSTER_LIST**: RR adds its cluster-ID; an RR ignores a route already containing its own cluster-ID.
- The RR does **not** change NEXT_HOP, AS_PATH, LOCAL_PREF or MED by default – clients must reach the original next-hop.
- Side effect: RR reflects only its own best path, hiding alternatives → use ADD-PATH or different RDs (VPN) for multipath and fast convergence.

Q8. What is a confederation and how does AS_PATH behave?

- The AS is split into sub-ASes (often private numbers); between sub-ASes the session is "confederation eBGP", inside each it is iBGP.
- Sub-AS numbers go in **AS_CONFED_SEQUENCE**, which is used for loops but **not counted** in path length and is stripped when leaving the confederation. NEXT_HOP, LOCAL_PREF and MED are preserved across sub-ASes.

3. Next-hop and validity

Q9 ★. R1 learns a route from eBGP and sends it to iBGP peer R2. R2 shows it but not as best ("inaccessible"). Why?

- eBGP→iBGP: NEXT_HOP is **unchanged** – R2 sees the ISP's link address.
- If that link is not in the IGP, R2 cannot resolve the next-hop → route invalid → not best → not installed or advertised.
- Fix: **next-hop-self** on R1 toward iBGP peers, or advertise the external link into the IGP (passive interface).
- Check: show bgp <prefix> (next-hop, "inaccessible"), show ip route <next-hop>.

Q10. A route is in the BGP table but not advertised to a neighbour. Checklist?

- Not the **best** path (only best is sent).
- Next-hop unreachable → not valid.
- iBGP split horizon (learned from iBGP, peer is iBGP, no RR).
- Outbound route-map / prefix-list / filter-list.
- **no-export** / **no-advertise** community.
- Not sent to the peer it came from (and eBGP peer would drop it anyway – AS loop).
- Address family not activated for that neighbour, or send-community missing (VPN/EVPN).
- Check: show bgp neighbor <x> advertised-routes, show bgp <prefix> (shows who it's advertised to).

Q11. What is RIB-failure?

- BGP chose a best path but couldn't install it in the RIB, usually because a route with better **administrative distance** (static, OSPF, connected) already exists. By default the route is still advertised to peers.

Q12. "network 10.1.0.0 mask 255.255.0.0" is configured but nothing is advertised. Why?

- The network statement needs an **exact match** (prefix and mask) in the RIB. If only 10.1.1.0/24 exists, add a static 10.1.0.0/16 to Null0 or use aggregate-address.

4. Best-path selection

Q13. Give the best-path order.

#	Attribute	Prefer
1	Weight (Cisco-local)	Highest
2	LOCAL_PREF	Highest (default 100)
3	Locally originated	network / aggregate / redistribute
4	AS_PATH length	Shortest
5	ORIGIN	IGP < EGP < incomplete
6	MED	Lowest (same neighbour AS by default)
7	eBGP vs iBGP	eBGP
8	IGP metric to next-hop	Lowest
9	Multipath / oldest eBGP path	Tie handling
10	Router-ID (ORIGINATOR_ID)	Lowest
11	CLUSTER_LIST length	Shortest
12	Neighbour address	Lowest

Q14. LOCAL_PREF vs MED vs AS_PATH prepend – which direction of traffic does each control?

Tool	Scope	Influences
Weight	One router only	That router's outbound choice
LOCAL_PREF	Whole AS (iBGP)	**Outbound** traffic – which exit we use
MED	Sent to neighbour AS	**Inbound** – which of our links they use (only if they honour it)
AS_PATH prepend	Global	**Inbound** – makes a path look longer
Communities	Agreed with ISP	Inbound – ISP sets local-pref based on our community

Q15. When are MEDs compared? What is deterministic-MED?

- By default only between paths from the **same neighbouring AS**; always-compare-med compares across ASes.
- Missing MED = 0 (best) by default; bgp bestpath med missing-as-worst treats it as worst.
- Without deterministic-MED, the result can depend on arrival order; deterministic-med groups paths by neighbour AS first, giving consistent results.

5. AS_PATH and communities

Q16 ★. Customer has two sites in AS 65001 connected to your PEs in AS 100. Site B never learns Site A's routes. Why and fix?

- Site B receives a route with AS_PATH "100 65001" – its own AS is in the path → eBGP **loop prevention** drops it.
- Fix on the PE: **as-override** (replaces 65001 with 100) or on the CE: **allowas-in**.
- With as-override, use SoO (Site of Origin) to stop loops when a site is multihomed.

Q17. Other AS_PATH facts interviewers probe?

- remove-private-as strips 64512–65534 (and 4200000000+ for 4-byte) before sending to eBGP.

- AS_SET (from aggregation with as-set) counts as 1 hop.
- 4-byte ASNs: AS_TRANS 23456 and AS4_PATH for old speakers.
- Prepend is applied outbound by you; local-as / replace-as for migrations.

Q18. Well-known communities and extended communities?

Community	Effect
NO_EXPORT	Do not send outside the AS (or confederation)
NO_ADVERTISE	Do not send to any peer
LOCAL_AS (NO_EXPORT_SUBCONFED)	Do not send outside the sub-AS
Extended: Route Target	VPN/EVPN import/export
Extended: SoO	Site of Origin loop prevention
EVPN ext. communities	MAC Mobility (sequence), Router MAC, ES-Import, Encapsulation (VXLAN)
Large communities	12-byte (ASN:x:y) for 4-byte ASNs

- Communities are not sent unless send-community (standard/extended/both) is configured.

6. Operations and convergence

Q19. Soft reconfiguration inbound vs route refresh?

- Soft-reconfig inbound keeps an unmodified copy of everything received (memory cost) so policy changes can be re-applied without asking the peer.
- Route refresh (capability in OPEN) asks the peer to resend; no extra memory. Preferred today. Hard reset (clear bgp *) tears down the session – avoid.

Q20. Graceful restart vs BFD – aren't they opposites?

- **Graceful restart**: during a control-plane restart, the peer keeps forwarding using stale routes (forwarding plane still works). Good for planned restarts/ISSU.
- **BFD**: detects forwarding failure in milliseconds and tears the session down fast. Good for link/path failure.
- Together: BFD-detected failure should override GR (the forwarding plane really failed).

Q21. What are ADD-PATH, max-prefix and dampening for?

- **ADD-PATH**: advertise multiple paths per prefix (RRs) → multipath, faster convergence, PIC edge.
- **maximum-prefix**: protect against route leaks; warn or tear down at the limit.
- **Dampening**: suppress a flapping prefix using penalties (default 1000 per flap, suppress 2000, reuse 750, half-life 15 min).

Q22. MP-BGP: how do VPN and EVPN routes ride BGP?

- Capabilities in OPEN negotiate AFI/SAFI: IPv4 unicast 1/1, VPNv4 1/128, VPNv6 2/128, L2VPN EVPN **25/70**, BGP-LU 1/4.
- NLRI and next-hop are carried in **MP_REACH_NLRI** / MP_UNREACH_NLRI attributes. Each AF must be activated per neighbour (address-family l2vpn evpn → neighbor x activate).

7. Testing angle (QA role)

If asked "How would you test BGP?" cover:

Area	Test ideas
Session	FSM states, timers negotiation, MD5, multihop, update-source, wrong AS (NOTIFICATION codes), max-prefix

Area	Test ideas
Propagation	iBGP split horizon, RR client/non-client rules, ORIGINATOR_ID/CLUSTER_LIST loops, confederations
Attributes	Each best-path step isolated; next-hop-self; LOCAL_PREF stripping to eBGP; MED rules; communities incl. no-export
Policy	Route-maps in/out, prefix-lists, AS-path filters, soft reconfig vs route refresh
Convergence / HA	Link/peer failure with BFD, GR/NSR, ADD-PATH/PIC, measured loss in ms
Scale / stress	Full Internet table (~1M IPv4), many peers, churn and flaps, memory/CPU
Interop	Against Cisco/Juniper/Nokia: capabilities, 4-byte ASN, EVPN/VPN AFI